G H Raisoni International Skill Tech University, Pune

End Semester Examinations
School of Engineering and Technology
FY B Tech, Semester - II (ALL)

Course Name: Programming Logic Skills **Course Code: 24BTET205**
Max Marks: 60 **Duration: 180 minutes**

Instructions:

1. Write the solution in detail with example.
2. Read each question carefully before attempting to answer.
3. Students must label their answers with the corresponding question number, such as Q. No. 1, Q. No. 2 etc.

Q. No.	Sub Q.No.	Question	Marks	CO#	BL
Q.1	Solve any Two				
	A	Explain the quantifiers and logical connectives in first order logic.	6	1	3
	B	Explain Demorgans law with example.	6	1	2
	C	Write the symbolic representations and truth table of the following connectives: Conjunction, Disjunction, Negation, Exclusive OR	6	1	2
Q.2	Solve any Two				
	A	Simplify the following **Canonical SOP expressions** into a **simplified SOP form**: a) F (A, B, C) = ∑m (1, 2, 3, 5, 6, 7) b) F (A, B, C, D) = ∑m (0, 1, 2, 4, 6, 8, 10)	6	2	2
	B	Explain the following gates with functioning, diagram, truth table. a) NAND b) EXOR	6	2	1
	C	Explain sequential circuit with its types.	6	2	2
Q.3	Solve any Two				
	A	Explain set, discuss its types with example.	6	3	2
	B	Discuss operations on set.	6	3	3
	C	Explain function and types of functions	6	3	4
Q.4	Solve any Two				
	A	Explain functional programming language, discuss characteristics, advantages and disadvantages.	6	4	1
	B	Differentiate between static and dynamic semantic.	6	4	2
	C	Explain why algorithm design and flowchart creation is important step in program development life cycle.	6	4	4
Q.5	Solve any Two				
	A	Explain algorithm and explain the properties that good algorithm must satisfy.	6	5	1
	B	Explain program tracing, purpose and when it is used.	6	5	2

	C	Explain time and space complexity. Find the time and space complexity of the following pseudo code.
		`def create_matrix(n):`
		`  matrix = []`
		`  for i in range(n):`
		`  row = [0] * n`
		`  matrix.append(row)`
		`  return matrix`
			6	5	3

- End of the Question Paper -